

# VAE-DSAA Module

## Technical Specification & JSON Templates

*Member 2 Output to Member 4 (Fusion Engine)*

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### 1. VAE-DSAA Output JSON Schema (Standard)

This is the formal structure Member 4 expects from the `/api/v1/behavioral/classify` endpoint. The VAE-DSAA module exposes a behavioral risk score derived from Dual-Signal Anomaly Attribution (reconstruction error + KL divergence decomposition) alongside a fraud typology label discovered via DBSCAN clustering.

```
{
  "transaction_ref": {
    "nameOrig": "string (Account ID)",
    "step": "integer (Simulation Hour)",
    "composite_id": "string (nameOrig_step)"
  },
  "behavioral_risk_score": "float (0.0 - 1.0)",
  "risk_level": "string (NORMAL | SUSPICIOUS | CRITICAL)",
  "detection_method": "Stratified-VAE-DSAA",
  "modality": "behavioral_anomaly",
  "transaction_type": "string (TRANSFER | CASH_OUT | PAYMENT)",

  "evidence": {
    "current_transaction": {
      "type": "string",
      "amount": "float",
      "F2_amount_to_balance_ratio": "float",
      "F3_balance_consistency_score": "integer (0|1)",
      "F4_balance_change_ratio": "float",
      "F5_dest_balance_ratio": "float",
      "F8_is_large_transaction": "integer (0|1)",
      "fraud_signal_summary": "string (Human-readable summary for LLM)"
    }
  },

  "anomaly_fingerprint": {
    "signal_1_reconstruction_error": {
      "description": "Per-feature % contribution to total reconstruction error",
      "F1_log_amount": "float",
      "F2_amount_to_balance_ratio": "float",
      "F3_balance_consistency_score": "float",
      "F4_balance_change_ratio": "float",

```

```

    "F5_dest_balance_ratio": "float",
    "F6_hour_of_day": "float",
    "F7_day_of_simulation": "float",
    "F8_is_large_transaction": "float",
    "dominant_feature": "string",
    "dominant_feature_signal": "string"
  },
  "signal_2_kl_divergence": {
    "description": "Per-latent-dim % contribution to total KL divergence",
    "dim_0": "float", "dim_1": "float", "dim_2": "float", "dim_3": "float",
    "dim_4": "float", "dim_5": "float", "dim_6": "float", "dim_7": "float",
    "dominant_dimension": "string",
    "dominant_dimension_signal": "string"
  }
},

"fraud_typology": {
  "cluster_id": "integer",
  "typology_label": "string (e.g. ACCOUNT_DRAINING | SMURFING_ESCALATION)",
  "typology_description": "string",
  "discovery_method": "DBSCAN"
},

"vae_diagnostics": {
  "vae_model_used": "string (VAE_TRANSFER | VAE_CASH_OUT | VAE_PAYMENT)",
  "reconstruction_error_total": "float",
  "kl_divergence_total": "float",
  "combined_anomaly_score": "float",
  "per_stratum_threshold": "float",
  "threshold_exceeded_by": "float"
}
}

```

## 2. Mock Data Generation Templates (For Member 4 Training)

Member 4 should use these two primary scenarios to train their ensemble meta-classifier (Logistic Regression stacking layer) and to test their RAG-grounded forensic report generator.

### Scenario A: Account Draining (High-Risk Fraud — CRITICAL)

- Logic: The current TRANSFER transaction near-fully empties the originator account into a previously-empty destination account. Balance consistency check (F3) fails, indicating a mathematical inconsistency characteristic of PaySim fraud.
- Target Scores: behavioral\_risk\_score > 0.90; risk\_level: CRITICAL
- DSAA Signal: signal\_1\_reconstruction\_error should assign dominant weight to F2\_amount\_to\_balance\_ratio (e.g., 0.41) and F3\_balance\_consistency\_score (e.g., 0.27). signal\_2\_kl\_divergence should show high loading on dim\_0 (balance-draining latent dimension).
- LLM Anchor — fraud\_signal\_summary: "Account fully drained into a previously empty destination account. Balance consistency check failed."
- FATF Typology Match: ACCOUNT\_DRAINING → maps to Layering / Cash-Out phase in Member 4's ChromaDB vector store.

```

{
  "transaction_ref": {
    "nameOrig": "C1231006815",
    "step": 206,
    "composite_id": "C1231006815_206"
  },
  "behavioral_risk_score": 0.94,
  "risk_level": "CRITICAL",
  "detection_method": "Stratified-VAE-DSAA",
  "modality": "behavioral_anomaly",
  "transaction_type": "TRANSFER",
  "evidence": {
    "current_transaction": {
      "type": "TRANSFER",
      "amount": 181234.56,
      "F2_amount_to_balance_ratio": 0.98,
      "F3_balance_consistency_score": 0,
      "F5_dest_balance_ratio": 1.0,
      "F8_is_large_transaction": 1,
      "fraud_signal_summary": "Account fully drained into a previously empty
        destination account. Balance consistency check failed. Transaction
        exceeds 95th percentile for TRANSFER type."
    }
  },
  "anomaly_fingerprint": {
    "signal_1_reconstruction_error": {
      "F1_log_amount": 0.08, "F2_amount_to_balance_ratio": 0.41,
      "F3_balance_consistency_score": 0.27, "F4_balance_change_ratio": 0.12,
      "F5_dest_balance_ratio": 0.07, "F6_hour_of_day": 0.02,
      "F7_day_of_simulation": 0.01, "F8_is_large_transaction": 0.02,
      "dominant_feature": "F2_amount_to_balance_ratio",
      "dominant_feature_signal": "Amount-to-balance ratio of 0.98 indicates
        near-total account draining - primary behavioral anomaly trigger."
    },
    "signal_2_kl_divergence": {
      "dim_0": 0.31, "dim_1": 0.09, "dim_2": 0.24, "dim_3": 0.05,
      "dim_4": 0.18, "dim_5": 0.04, "dim_6": 0.06, "dim_7": 0.03,
      "dominant_dimension": "dim_0",
      "dominant_dimension_signal": "Latent dim_0 encodes balance-draining
        behavioral patterns for TRANSFER type - high KL deviation confirms
        account-emptying fraud signature."
    }
  },
  "fraud_typology": {
    "cluster_id": 2,
    "typology_label": "ACCOUNT_DRAINING",
    "typology_description": "Full or near-full balance extraction via TRANSFER
      into a fresh destination account.",
    "discovery_method": "DBSCAN"
  },
  "vae_diagnostics": {

```

```

    "vae_model_used": "VAE_TRANSFER",
    "reconstruction_error_total": 0.873,
    "kl_divergence_total": 1.204,
    "combined_anomaly_score": 0.94,
    "per_stratum_threshold": 0.61,
    "threshold_exceeded_by": 0.33
  }
}

```

## Scenario B: Smurfing / Escalating Rhythm (SUSPICIOUS)

- Logic: A sequence of incrementally increasing CASH\_OUT amounts from the same originator, deviating from the normal behavioral cadence learned by VAE\_CASH\_OUT.
- Target Scores: behavioral\_risk\_score: 0.45 – 0.70; risk\_level: SUSPICIOUS
- DSAA Signal: signal\_1\_reconstruction\_error dominant on F4\_balance\_change\_ratio and F6\_hour\_of\_day (escalating rhythm + temporal anomaly). signal\_2\_kl\_divergence dominant on dim\_4 (rhythm/cadence latent dimension for CASH\_OUT type).
- LLM Anchor — fraud\_signal\_summary: "Incrementally escalating CASH\_OUT amounts detected — consistent with smurfing escalation rhythm."
- FATF Typology Match: SMURFING\_ESCALATION → maps to Structuring / Smurfing in Member 4's ChromaDB vector store.

```

{
  "transaction_ref": {
    "nameOrig": "C8821334420",
    "step": 314,
    "composite_id": "C8821334420_314"
  },
  "behavioral_risk_score": 0.58,
  "risk_level": "SUSPICIOUS",
  "detection_method": "Stratified-VAE-DSAA",
  "modality": "behavioral_anomaly",
  "transaction_type": "CASH_OUT",
  "evidence": {
    "current_transaction": {
      "type": "CASH_OUT",
      "amount": 9400.00,
      "F2_amount_to_balance_ratio": 0.34,
      "F3_balance_consistency_score": 1,
      "F5_dest_balance_ratio": 0.12,
      "F8_is_large_transaction": 0,
      "fraud_signal_summary": "Incrementally escalating CASH_OUT amounts detected. Current amount 22% larger than preceding transaction by same originator - consistent with smurfing escalation rhythm."
    }
  },
  "anomaly_fingerprint": {
    "signal_1_reconstruction_error": {
      "F1_log_amount": 0.05, "F2_amount_to_balance_ratio": 0.19,
      "F3_balance_consistency_score": 0.03, "F4_balance_change_ratio": 0.31,
      "F5_dest_balance_ratio": 0.06, "F6_hour_of_day": 0.28,
      "F7_day_of_simulation": 0.04, "F8_is_large_transaction": 0.04,
    }
  }
}

```

```

    "dominant_feature": "F4_balance_change_ratio",
    "dominant_feature_signal": "Proportional balance impact escalating
      across successive transactions - indicative of smurfing pattern."
  },
  "signal_2_kl_divergence": {
    "dim_0": 0.07, "dim_1": 0.11, "dim_2": 0.08, "dim_3": 0.14,
    "dim_4": 0.38, "dim_5": 0.09, "dim_6": 0.08, "dim_7": 0.05,
    "dominant_dimension": "dim_4",
    "dominant_dimension_signal": "Latent dim_4 encodes transaction rhythm
      for CASH_OUT type - elevated KL suggests deviation from normal cadence."
  }
},
"fraud_typology": {
  "cluster_id": 5,
  "typology_label": "SMURFING_ESCALATION",
  "typology_description": "Sequence of incrementally increasing transactions
    designed to stay below reporting thresholds while consolidating funds.",
  "discovery_method": "DBSCAN"
},
"vae_diagnostics": {
  "vae_model_used": "VAE_CASH_OUT",
  "reconstruction_error_total": 0.412,
  "kl_divergence_total": 0.687,
  "combined_anomaly_score": 0.58,
  "per_stratum_threshold": 0.49,
  "threshold_exceeded_by": 0.09
}
}

```

### 3. Critical Factual Anchors for Member 4

For the LLM to generate forensic reports without hallucinating, these fields are mandatory in every VAE-DSAA output and must be injected into the Chain-of-Evidence prompt template:

| Field                     | Purpose for Member 4                                                                                                                                                                        |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| composite_id              | Primary key (format: nameOrig_step). Used by Member 4 to link the behavioral signal with Member 1's Graph data and Member 3's TS-TCN temporal signal. All three modalities share this key.  |
| fraud_signal_summary      | Provides the 'Behavioral Reasoning' anchor for the LLM forensic report. Injected directly into the Chain-of-Evidence prompt to ground the narrative in VAE-observed behavioral evidence.    |
| dominant_feature_signal   | Explains which specific behavioral feature (e.g., F2_amount_to_balance_ratio) drove the reconstruction anomaly. Gives Member 4's LLM feature-level forensic attribution without SHAP.       |
| dominant_dimension_signal | Explains which latent behavioral pattern (e.g., dim_0 = balance-draining pattern) was violated in the KL space. Provides the second layer of forensic attribution unique to the VAE module. |

|                       |                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| typology_label        | DBSCAN-discovered fraud category (e.g., ACCOUNT_DRAINING, SMURFING_ESCALATION). Used as the semantic search query into Member 4's ChromaDB FATF typology vector store for RAG retrieval.          |
| behavioral_risk_score | The primary numeric input to Member 4's Logistic Regression meta-classifier, alongside Member 1's graph score and Member 3's temporal_risk_score. Float 0.0–1.0.                                  |
| vae_model_used        | Identifies which stratified VAE processed this transaction (VAE_TRANSFER, VAE_CASH_OUT, VAE_PAYMENT). Allows Member 4's report to specify which type-specific behavioral model flagged the alert. |

## 4. Integration Details

| Parameter                      | Specification                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Endpoint                       | POST /api/v1/behavioral/classify                                                                                                                               |
| Framework                      | FastAPI with Pydantic schema validation enforcing the JSON contract above                                                                                      |
| Performance Requirement        | Response within 50ms per transaction (individual inference). Compatible with Member 4's 500ms orchestration timeout.                                           |
| Shared Key                     | composite_id (nameOrig_step) — must match the format used by Member 1 (graph nodes) and Member 3 (TS-TCN transaction_ref.composite_id)                         |
| VAE Models                     | Three separate models: VAE_TRANSFER, VAE_CASH_OUT, VAE_PAYMENT. Routing is handled internally by the stratification layer before Member 4 receives the output. |
| Anomaly Fingerprint Dimensions | Signal 1: 8 values (one per feature F1–F8, summing to 1.0). Signal 2: 8 values (one per latent dimension dim_0–dim_7, summing to 1.0).                         |
| Risk Levels                    | NORMAL: combined_anomaly_score <= per_stratum_threshold.<br>SUSPICIOUS: exceeds threshold by < 0.15. CRITICAL: exceeds threshold by >= 0.15.                   |
| Visual Content for Member 4    | Provide mock ROC curve data and ablation study results comparing stratified vs. global VAE configurations to populate the forensic dashboard visual sections.  |

## 5. DSAA Dual-Signal Architecture Summary

The VAE-DSAA module produces two independent anomaly signals that together form the Anomaly Fingerprint. This two-layer structure is what makes the behavioral modality unique within the DeepSentinel platform and distinguishes it from a simple anomaly score.

| Signal | What it measures | Member 4 usage |
|--------|------------------|----------------|
|--------|------------------|----------------|

|                                 |                                                                                                                                                                                         |                                                                                                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal 1 — Reconstruction Error | Per-feature percentage contribution to total reconstruction error. Reveals which transaction characteristics (F1–F8) deviated from the type-specific normal pattern learned by the VAE. | dominant_feature_signal is injected into the Chain-of-Evidence prompt. Tells the LLM which behavioral feature drove the behavioral alert.                                              |
| Signal 2 — KL Divergence        | Per-latent-dimension percentage contribution to total KL divergence. Reveals which learned behavioral patterns in the latent space (dim_0–dim_7) were violated by this transaction.     | dominant_dimension_signal provides a second forensic attribution layer. Together with Signal 1, it constitutes the full Anomaly Fingerprint vector used for DBSCAN typology discovery. |